Ericsson PE (HEB-PE-12) Optical Transport Failure SOP (ALARM_4441)

Node: HEB-PE-12 | **Vendor:** Ericsson | **Severity:** Minor | **Alarm Code:** ALARM_4441

Description

This standard operating procedure addresses a minor optical transport failure on the Ericsson PE (HEB-PE-12) at the Hebbal Edge site. The failure is attributed to SFP/XFP degradation, fiber cuts, or Rx/Tx power loss, which can be caused by a combination of hardware and software issues. The Ericsson PE, being an Optical Transport Network (OTN) node, relies on the SFP/XFP modules to transmit and receive optical signals over the fiber links. When a failure occurs, the node may not be able to maintain the required signal quality, leading to signal degradation, errors, or even link loss. The OTN protocol, which is used for transporting client signals over the optical links, is based on the Generalized Multi-Protocol Label Switching (GMPLS) architecture. GMPLS uses a label switched path (LSP) to establish a connection between two nodes, and the OTN header is used to carry the client signal. In the event of a failure, the OTN header may not be properly processed, leading to errors and alarms. This SOP outlines the steps to diagnose and resolve the failure, ensuring minimal disruption to the network.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Hebbal Edge site to provide physical access to the equipment.

- Verify the status of the redundant link to ensure it is available and can take over in case of a failure.

- Check the environmental conditions at the site, including temperature, humidity, and power supply, to ensure they are within the recommended operating range.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: admin show interfaces , Expected Output: Interface name, status, and signal quality metrics, Description: This command is used to verify the status of the interface and check for any signal quality issues.

Command: admin show optical signals , Expected Output: Optical signal power, wavelength, and other relevant metrics, Description: This
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command is used to check the optical signal quality and identify any issues.

Command: admin show sfp , Expected Output: SFP/XFP module details, including temperature and voltage, Description: This command is used to verify the SFP/XFP module details and check for any issues.
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Resolution Steps

Step 1: Step: Swap the SFP/XFP module, Command: admin replace sfp , Description: Swap the SFP/XFP module with a new one to resolve the issue. Ensure the new module is compatible with the existing equipment.

Step 2: Step: Adjust the optical signal power, Command: admin adjust optical power , Description: Adjust the optical signal power to ensure it is within the recommended range. This may involve adjusting the transmitter power or the receiver sensitivity.

Step 3: Step: Check and repair fiber cuts, Command: , Description: Check the fiber links for any cuts or damage and repair them as necessary. This may involve dispatching a fiber technician to the site.

Post-Check Verification

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Command: admin show interfaces , Expected Output: Interface name, status, and signal quality metrics, Description: Verify the status of the interface and check for any signal quality issues.

Command: admin show optical signals , Expected Output: Optical signal power, wavelength, and other relevant metrics, Description: Verify the optical signal quality and check for any issues.

Command: admin show sfp , Expected Output: SFP/XFP module details, including temperature and voltage, Description: Verify the SFP/XFP module details and check for any issues.
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Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: admin replace sfp , Description: Restore the original SFP/XFP module if the new one does not resolve the issue.

- Step: Revert the optical signal power adjustment, Command: admin adjust optical power , Description: Revert the optical signal power adjustment if it does not resolve the issue.

- Step: Re-check and repair fiber cuts, Command: , Description: Re-check the fiber links for any cuts or damage and repair them as necessary if the issue persists.